Move DB (database Query and update on GPU)

- ✓ select or search query has been addressed by community.
- ✓ we can do update on GPU.

E.g. We have two tables, A and B.

- ✓ for each record a_i in A, we search in table B, Then updata a_i;
- ✓ This is a $O(n^2)$ algorithm, Without other data structures.
- ✓ If size of (A, B) is one million or more, it takes hours for this update operation!

IDEA of project:

1. we Load table data A and B onto GPU memory!
2. Convert SQL statement into C or C++ function calls with parameters. (This is standard Library in open source DB, such as sqlite)
3. Use GPU to perform Query and update in parallel. tables A or B is considered as 2D array!
4. For simplicity, we first focus on medium-sized data set, table A < 1GB, which can be fitt into memory! Later, we address bigger dataset that greater than GPU memory capacity!
5. We can use MongoDB or sqlite for test!
6. explore new cuda features that support these operations such as cuda stream or Thrust!

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⑦ explore data organization on GPU: its impacts on Query and update performance! E.g. B tree on GPU!

— B tree on GPU

— R tree on GPU

— compression / decompression of the data on GPU
2D array compression lossless!

— divide Large big dataset into Chunks, Load each Chunk onto GPU in turns!

⑧ explore These Ideas on Mongo DB, open source & implemented ~~on~~ with C++;